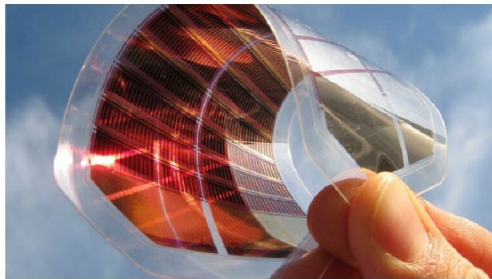


Organic solar cells reach record efficiency



Organic solar cell (representational image)

University of Michigan researchers have demonstrated organic solar cells that can achieve 15 per cent efficiency, in an advance that makes a more flexible, inexpensive type of solar cell commercially viable. This level of efficiency is in the range of many solar panels, or photovoltaics, currently on the market.

According to Stephen Forrest, the Peter A. Franken Distinguished University professor of engineering, and Paul G.

Goebel, professor of engineering, who led the work, "Organic photovoltaics can potentially cut way down on the total solar energy system cost, making solar a truly ubiquitous clean energy source."

At 15 per cent efficiency and a 20-year lifetime, researchers estimate organic solar cells could produce electricity at a cost of less than 7 cents per kilowatt-hour. In comparison, the average cost of electricity in the US was 10.5 cents per kilowatt-hour in 2017, according to US Energy Information Administration.

As per the university's press release, organic solar cells incorporate carbon into their construction to offer several advantages over conventional inorganic cells. Silicon-based inorganic solar panels are costly to make, since these are composed of thick, rigid sheets that require fixed installation points. However, carbon-based organic solar cells could be inexpensively manufactured in rolls that are thin enough to bend and curve around structures or within clothing, and made in any colour, even transparent, to blend in with their environment.

World's fastest drone by Zipline

Zipline launches the fastest commercial delivery drone on the planet. The Silicon Valley-based company's mission is to bring life-saving medicines to the most hard-to-reach places on Earth. They operate the world's first and only national-scale drone delivery system, making thousands of deliveries per year.

The aircraft is part of a complete redesign of Zipline's logistics system, which dramatically improves the system's launch, autonomous flight and landing capabilities. Improvements due to the redesign will decrease the amount of time between Zipline's receipt of an order and launch of a fulfilment flight from ten minutes to one, increase the number of daily delivery flights that each Zipline distribution centre can make from 50 to 500 and expand the radius of each distribution centre to serve populations of up to 10 million people.

Zipline's new delivery vehicle is an autonomous, fixed-wing style airplane, which can fly at a top speed of 128kmph and a cruising speed of 101kmph—21 kmph faster than the previous generation of aircraft—with a round trip range of 160 kilometres, carrying up to 1.75 kilos of cargo. It can fly



The aircraft flies farther, faster and with more cargo than ever before, come high altitude, heavy wind or rain (Credit: <https://dronedj.com>)

four times faster than the average quadcopter drone and can serve an area 200 times as large.